# From questions to insights: a reproducible question-answering pipeline for historiographical corpus exploration

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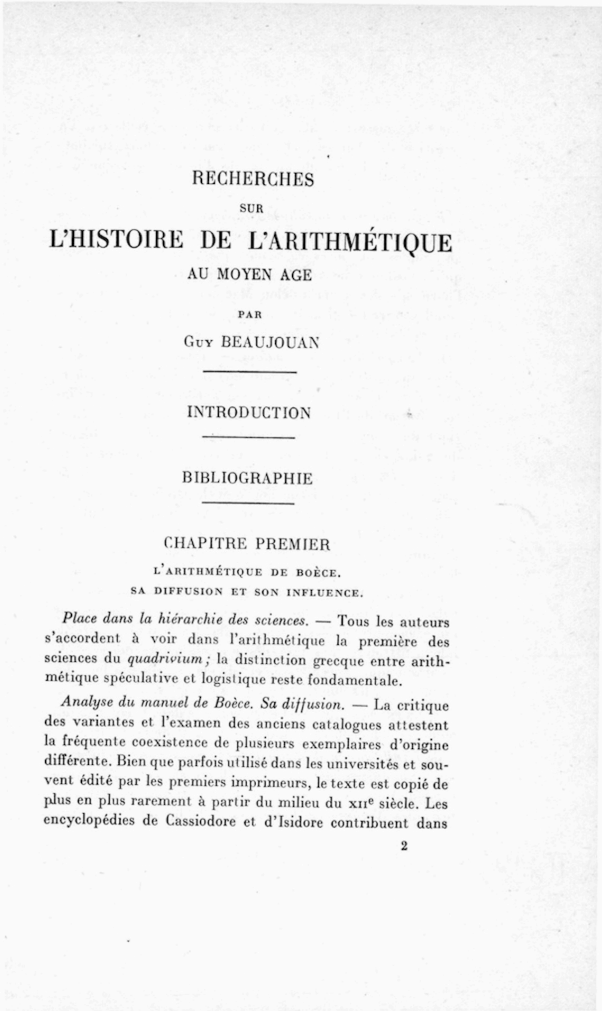
## Context

Since 1849, the École des Chartes-PSL has been publishing longform abstracts of institutional theses defended by its students (Guyotjeannin 1997). It is an online corpus of over 3,000 scientific texts, in French, covering the transcription, criticism and diplomatic editing of ancient texts. This is a rare witness of the evolution of historical disciplines since the second half of the 19th century, and some of its works are pioneers in the renewal of analytical methods in history (Figure 1). However, this corpus is actually underexploited by researchers due to limited search capabilities.



RECHERCHES
SUR
L'HISTOIRE DE L'ARITHMÉTIQUE
AU MOYEN AGE
PAR
GUY BEAUJOUAN

INTRODUCTION

BIBLIOGRAPHIE

CHAPITRE PREMIER
L'ARITHMÉTIQUE DE BOÈCE.
SA DIFFUSION ET SON INFLUENCE.

*Place dans la hiérarchie des sciences.* — Tous les auteurs s'accordent à voir dans l'arithmétique la première des sciences du *quadrivium;* la distinction grecque entre arithmétique spéculative et logistique reste fondamentale.

*Analyse du manuel de Boèce. Sa diffusion.* — La critique des variantes et l'examen des anciens catalogues attestent la fréquente coexistence de plusieurs exemplaires d'origine différente. Bien que parfois utilisé dans les universités et souvent édité par les premiers imprimeurs, le texte est copié de plus en plus rarement à partir du milieu du XIIe siècle. Les encyclopédies de Cassiodore et d'Isidore contribuent dans

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*Longform thesis abstract (called "positions de thèses") first page exemple. Guy Beaujouan, « Recherches sur l'histoire de l'arithmétique au Moyen Âge », in Positions des thèses soutenues par les élèves de la promotion de 1947 pour obtenir le diplôme d'archiviste paléographe, École nationale des chartes, Paris, 1947, p.17-22.*

## Issues. Limited search capabilities

Full-text searching, only based on term occurrences, involves browsing a large collection of "noisy" results before identifying the most relevant document. Furthermore, this type of search does not allow the discovery of documents based on their semantic or thematic similarity.

# State of the art. Opportunities and limitations

Information retrieval (IR) systems, which typically rely on proximity measure applied to pre-trained vector representations (also called "embeddings"), prove to be effective in retrieving the most relevant documents according to their semantic proximity to a concept or sentence (top-k embedding-based lookup). These IR systems are generally based on hybrid combinations of embedding models such as Word2Vec or Doc2Vec (Mikolov et al. 2013; Le / Mikolov 2014) or even on transformer-type architectures (Vaswani et al. 2023). In Digital humanities, they facilitate for researchers the identification of specific concepts in complex textual collections (Camps et al. 2023) while offering powerful tools for document retrieval based on semantic proximity. More recently, Large Language Models (LLM), which are based on a prompt system and Retrieval-Augmented Generation (RAG) frameworks such as LangChain , offer more intuitive querying modalities, making it possible to personalize or synthesize output content. However, these models present some limitations applied to humanities: responses are too short and often decontextualized, "hallucinations" sometimes give the illusion of structured output, and finally, the lack of traceability is prohibitive from the educational point of view.

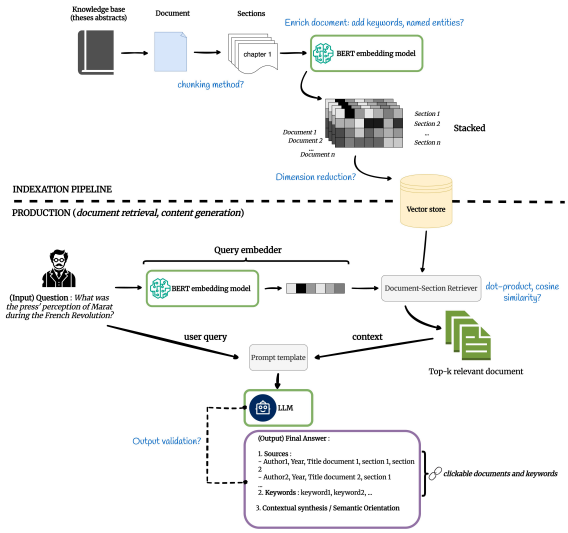
# Open question

How can these systems be adapted to domain-specific corpora? Is it possible to structure and contextualize (i.e. to provide sources) the answer to a user question? These tools offer researchers the opportunity to discover unsuspected connections between documents. For example, a seemingly simple query such as "What are popular musical practices in the cultural history of France?" could lead to unexpected abstracts: from medieval songs with troubadours , to Eurovision show history , to rock and pop broadcasts on French television .

# Objectives

The goal is not to develop a conversational bot, but to design a reproducible IR question answering (Q&A) pipeline (Figure 2) where each step is documented and adapted to our use case of theses abstracts. We aim to provide the user with a structured and sourced response to allow them to easily navigate the entire collection (i.e. bounces between documents). For that, we propose a response template that contains three main elements (Figure 3):

1. **Keywords:** extraction of the main topics of the identified sections, enabling navigation to other abstracts close by subject.
2. **Sources:** providing links to abstracts and their associated sections, relevant to the user question.
3. **Contextual synthesis and semantic orientation:** a natural language response that synthesizes the documents identified and retrieved relates to the user question. This content is also designed to spark the researcher's curiosity, suggesting new ways of interest.




*The Information Retrieval Question-Answering pipeline as document retrieval system to explore large corpora.*

# Methodology

The pipeline (Figure 2) designed to produce the final structured response consists into three main stages:

## Vector-Based corpus indexing

The first stage, conducted as part of a pre-production phase, involves generating embeddings to support downstream retrieval and content generation tasks. These embeddings capture both the internal structure of theses abstracts (e.g. introduction, chapters, appendices) and, optionally, a document-level representation (e.g. named entities, keywords). We use open-source pre-trained

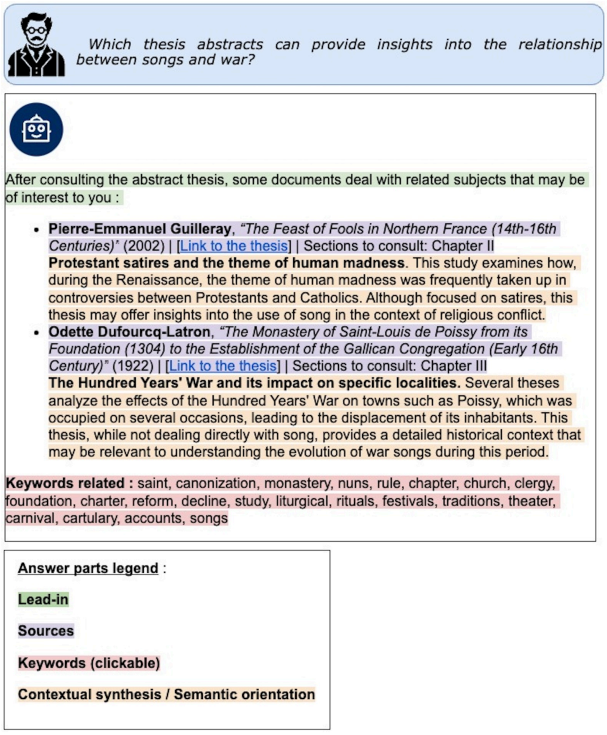
BERT-type sentence embedding models (Devlin et al. 2019; Reimers / Gurevych 2019), specifically fine-tuned and distilled to obtain these representations. A chunking method—segmenting abstracts into smaller, semantically coherent units—is considered to improve granularity and contextual alignment. To further enhance the expressiveness of the embeddings, we enrich them with metadata such as keywords (extracted using KeyBERT ) and named entities (identified with spaCy ), following a previous experiment on the same corpus (Jolivet / Torres Aguilar 2022). These representations, along with their associated metadata, are then indexed in a vector store (e.g. FAISS ; Douze et al. 2024), enabling efficient semantic retrieval. We also explore whether dimensionality reduction (e.g. principal component analysis — PCA) can optimize storage space within the vector store while preserving semantic expressiveness.

## Document-Retrieval module

During this second stage, the researcher expresses a query in natural language, which is converted into a dense vector using the same embedding model as in the corpus indexing stage. The system then compares the semantic representation of the query to those of the indexed documents in order to retrieve the most relevantdocuments based on semantic proximity (top-k most similar documents).

## Content generation and enrichment

The final experimental stage aims to post-process and aggregate document content into a coherent context in order to build a prompt and ultimately enrich the results using a large language model (LLM) to generate original content. This stage addresses two main issues: assessing the relevance and added value of the generated output in complementing the initial retrieval, and evaluating the performance of open-source models adapted to French (e.g. Mistral-7B or Mistral-Nemo ). It also examines the limitations of such systems, particularly constraints related to the maximum number of input tokens. Several optimizations are explored. The use of lightweight quantized models enables local execution in dedicated environments (e.g. LM Studio , Llama.cpp , Ollama ), thereby reducing reliance on paid external APIs while maintaining acceptable performance for simple tasks. Prompt engineering and structured output formatting—via JSON schemas—further contribute to the coherence and determinism of the generated content.




*Mock-up showing interaction between user and information retrieval system.*

# Conclusion

The Q&A pipeline we propose relies on a combination of advanced technologies to go beyond traditional documentary retrieval approaches in large corpora by integrating semantic and contextual dimensions. Based on BERT models for hybrid embedding, vector stores for indexing and semantic search, and LLM for content generation, our system aims to offer a more contextualized exploration of corpora. However, several technological and methodological limitations appear. The re-use of pre-trained models, although efficient, remains dependent on their fine-tuning to the specific domain. LLM, in particular, raise a lot of challenges too, which we are attempting to explore in this experiment. These limitations highlight the need to control each stage of this pipeline by combining optimization techniques (dimensional reduction, structured prompts, reranking and output validation) and validation methods to guarantee the quality and reliability of the answers produced. We explore the construction of dynamic, interconnected document corpora, offering researchers not only relevant documents, but also unexpected thematic links. Finally, this method, while not revolutionary, is a way of enhancing the access to the content of humanities textual databases.

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